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Perspectives on integrated continuous bioprocessing — opportunities and challenges

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Biopharmaceuticals are currently produced almost entirely using batch operations. Integrated continuous processes have the potential to revolutionize bioprocessing, leading to significant reductions in manufacturing costs and facility size while improving product quality through enhanced uniformity in the microenvironment. This paper examines the potential opportunities and challenges in implementing continuous processes for the production of high value biological products. Although regulatory agencies seem highly open to continuous processing, there are significant technical and practical issues that must be addressed to move the industry in this direction, including the development of effective alternatives to traditional downstream processing operations. Continuous processing could provide unique opportunities for the production and delivery of low-cost biopharmaceuticals for solution of major global health challenges in the coming decades.

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Introduction

The commercial-scale manufacture of high value biological products is currently performed using batch processes in which each unit operation is completed in sequence, with the product outflow from one unit typically collected in a holding tank before moving to the next processing step. Batch operation facilitates the design and optimization of the individual unit operations, often by individuals/teams with very different expertise (e.g., upstream vs downstream or membrane filtration vs chromatography). Batch processing is also attractive given the relatively small scales needed for most biological products, and it easily accommodates off-line measurements of key product quality attributes between processing steps.

The transition from batch to continuous processing has been a hallmark in the development of the modern chemical and pharmaceutical industries. For example, Kreps' early analysis of the history of sulfuric acid production noted that [1]:

> "Chemical research and technology gradually coalesce discontinuous processes into continuous ones, replace batch operations by complete unit operations. . .and thus reduce operating charges and increase productive capacity per unit of investment."

Similarly, soda ash production in the 18th and 19th centuries was dominated by the Leblanc process, which involved the batch reaction of sodium chloride with sulfuric acid to produce sodium sulfate which was then reacted with coal and calcium carbonate to form sodium carbonate (with calcium sulfide as a by-product). The Solvay process, developed in the 1860s by Ernest Solvay, was a continuous process in which carbon dioxide was reacted in an aqueous solution of NaCl and ammonia yielding sodium bicarbonate (with NH_4Cl as by-product) that was then converted to sodium carbonate by calcination [2]. Within 30 years, more than 90% of the global production of sodium carbonate was based on the continuous Solvay process; companies that had invested in the Leblanc process were almost entirely supplanted. As discussed by Stephanopoulos and Reklaitis [3], the Solvay process was one of the first examples highlighting the advantages of process systems engineering — it improved yields and productivity, reduced environmental pollution and manufacturing costs, and enhanced safety.

Drivers for continuous bioprocessing

There are two primary drivers behind the current interest in continuous bioprocessing. The first is economic. Biomanufacturers are faced with increasing pressure to reduce the price of biotherapeutics, particularly in light of efforts to increase the availability of life-saving therapies to patients in developing countries. In addition, there is growing competition from lower cost manufacturers, many of whom are focused on the production of biosimilars, that is, generic versions of biological molecules that are now off-patent. There is thus tremendous interest in finding ways to reduce the cost of all aspects of drug discovery, clinical trials, and manufacturing.

A recent economic analysis by Walthe *et al.* [4••] concluded that an integrated continuous biomanufacturing platform could reduce costs (net present value) by 55%

relative to conventional batch processing. Even greater benefits were found for non-monoclonal antibody products — the expected operating cost was reduced from $1230 g^{-1} for a batch process to less than $250 g^{-1} for a continuous process with more than a 3-fold reduction in capital costs. Similar cost-savings were identified by Hammerschmidt *et al.* [5] for production of a monoclonal antibody product using a continuous process based on precipitation instead of batch column chromatography. Continuous processes could also facilitate the design of flexible multi-product manufacturing facilities with much lower initial capital investments, which would allow biomanufacturers to more effectively manage rapid changes in product portfolios/demand and could be an enabling strategy for production of more personalized biotherapeutics.

In addition to cost, continuous processing also has the potential to provide significant improvements in product quality through enhanced control and uniformity of the microenvironment within the manufacturing process. Biotherapeutics produced using current batch processes have wide variability in 'experiences'. For example, a monoclonal antibody secreted by a Chinese Hamster Ovary (CHO) cell at the start of the cell culture is produced in a nutrient-rich environment with few lysed cells, but that protein will remain within the bioreactor for multiple days before subsequent downstream processing. The situation is dramatically different for a protein produced right before harvesting. Several studies have shown that much of the variability in glycosylation profiles [6], extent of deamidation [7], and level of degradation/aggregation [8] is due to the wide range (and very long) residence times inherent in batch processes. In addition, protein aggregation and denaturation can occur when proteins are bound to chromatographic resins for long times due to protein unfolding and intermolecular interactions involving other adsorbed proteins [9–11]. Proteins that are loaded on the column at the start of a typical column chromatography step remain in the bound state for well more than an hour, while proteins near the end of the load remain bound for only a small fraction of that time.

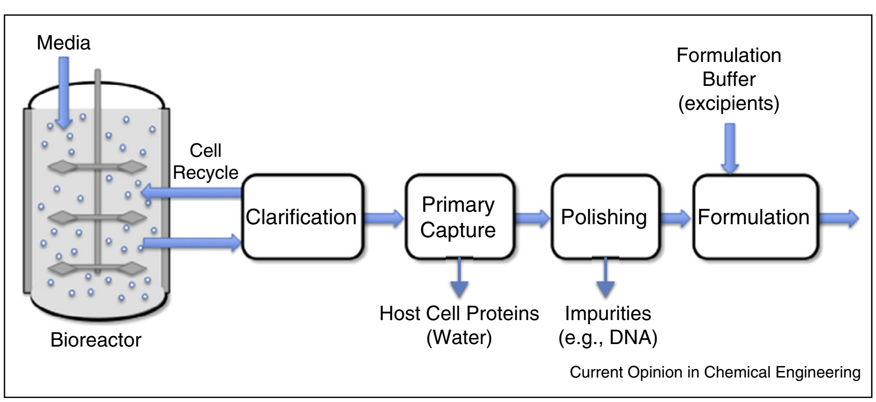
The FDA's recent Regulatory Science Strategic Plan [12] specifically focused on the use of Quality by Design (QbD) to improve the manufacturing process to ensure and improve product quality. As part of this effort, the FDA identified three new areas that would support increased manufacturing quality, one of which is the use of 'continuous processing where materials constantly flow in and out of equipment' [12]. Janet Woodcock, Director of the Center for Drug Evaluation and Research, recently identified continuous manufacturing as a key enabler in modernizing pharmaceutical manufacturing [13].

Technologies for continuous bioprocessing

There have been several recent Reviews of available technology for continuous bioprocessing [14,15••,16••]. Figure 1 shows a schematic of a generic continuous process that would be appropriate for purification of a secreted product like a monoclonal antibody (mAb). The overall process includes a perfusion bioreactor, clarification (with integrated cell recycle), initial product capture, product polishing, and final formulation. Additional steps (e.g., cell lysis) would be required for intracellular proteins or cellular products (e.g., viruses used in the formulation of vaccines).

The technology for continuous cell culture was developed more than 30 years ago [17]. The use of perfusion bioreactors with continuous addition of nutrients and removal of product is well-established for production of both monoclonal antibodies (mAb) and highly labile enzymes and cytokines [18]. Approximately 20 marketed monoclonal antibody products, with annual revenues around $20 billion, are currently produced using perfusion systems [19]. Voissard *et al.* [20] have reviewed available methods for cell retention in perfusion culture of mammalian cells, including filtration, centrifugation, and gravity settlers. The most common approaches are

Figure 1



Schematic representation of a generic continuous downstream bioprocess.
Source: Adapted from Ref. [16••].

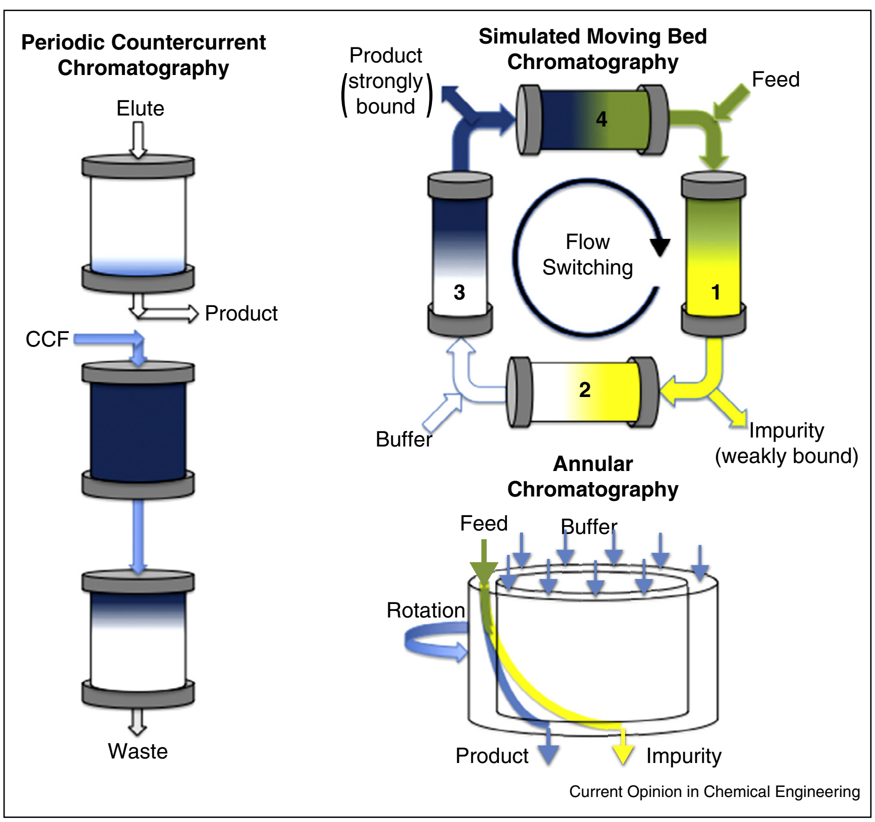
the use of spin-filters placed directly in the bioreactor, typically mounted as an annulus around the impeller shaft at the center of the bioreactor [21], and tangential flow filtration, with the cell separation unit located outside the bioreactor [22]. Alternating tangential flow filtration (ATF) has been specifically designed for use with perfusion bioreactors. ATF uses standard hollow fiber membrane modules with the flow driven back-and-forth through the fibers using a very low shear diaphragm pump that cyclically withdraws and returns the cell suspension to the bioreactor [23]. This minimizes cell damage, and the alternating flow reduces fouling and fiber clogging. One of the main advantages of an external filtration device is that the module can be readily replaced with minimal disruption to the bioreactor.

In contrast to perfusion bioreactors, the most appropriate technologies for continuous downstream processing are still a matter of considerable debate and ongoing development [16••,24]. In some cases, continuous operation can be achieved simply by using multiple disposable (single use) units, for example, one can perform clarification by operating two depth filters in parallel with the switch from filter A to filter B made at an appropriate end-point. On the other hand, initial protein purification (capture) is typically done using bind-and-elute column chromatography, for example, Protein A chromatography for monoclonal antibodies or ion exchange for many enzymes/cytokines. In each case the column is loaded, washed, and eluted in a fully sequential batch process (with the resin used for one operation before moving to the next).

A number of approaches have been developed to achieve continuous chromatographic separations (Figure 2), most of which are based on multi-column systems. This includes periodic counter-current chromatography (PCC) [25], simulated moving bed chromatography (SMB) [26•], and multicolumn countercurrent solvent gradient purification (MCGSP) [27]. These systems provide periodic (cyclical) operation, with the product concentration in the eluent stream varying with time during each cycle. Annular [28] and radial [29] chromatography can also be operated in continuous mode with true steady-state operation, although neither of these have been implemented for large-scale commercial protein purification.

Alternatively, continuous countercurrent tangential chromatography (CCTC) utilizes the chromatographic resin

Figure 2



Schematic of continuous chromatographic processes including periodic countercurrent chromatography, simulated moving bed chromatography, and annular chromatography.
Source: Adapted from Ref. [16••].

in the form of a slurry which flows sequentially through a series of static mixers and hollow fiber membrane modules [30•]. The micro-porous hollow fiber membranes retain the chromatographic resin while letting proteins and buffer components pass into the permeate. The buffers used in the binding, washing, elution, and regeneration steps flow countercurrent to the resin slurry in a multi-stage configuration within each step, similar to the use of countercurrent staging in classical mixer-settler systems for extraction (with the hollow fiber membrane replacing the settler for the 'phase' separation). In contrast to the periodic behavior of multi-column chromatography systems, CCTC provides truly continuous steady-state operation, with the product obtained from the elution step at constant concentration and quality due to the uniformity in residence time in the binding, washing, and elution stages [30•].

Although initial product capture is currently dominated by chromatographic separations, a number of alternative technologies could become attractive at the high product titers that can be achieved using optimized media and cell lines. These include protein precipitation [5] and crystallization [31], both of which tend to be more selective as the product titer increases, as well as aqueous two-phase extraction [32]. Although precipitation and crystallization are often thought of as batch operations, they can also be performed in a continuous mode using a tubular reactor consisting of a long tube with a static mixer insert to obtain uniform residence time and effective mixing of the reagents and of the formed precipitate [33].

Nearly all biotherapeutics are currently formulated using ultrafiltration (UF) and diafiltration (DF) to achieve the desired product concentration and buffer composition [34]. UF/DF processes are currently performed as batch operations. Continuous versions of these membrane processes could potentially be developed based on membrane cascades [35] and/or single pass tangential flow filtration [36], although additional work will be needed to demonstrate the effectiveness of these methods, particularly for highly concentrated monoclonal antibody formulations [16••]. Alternatively, product formulation could be achieved using lyophilization or crystallization in continuous tubular reactor systems [37].

In addition to the key product purification steps, providing high levels of viral clearance is a critical component of the downstream process for all biotherapeutics [38]. Current regulatory guidelines require the use and validation of at least two virus removal/inactivation steps, which must be based on 'orthogonal' mechanisms of virus clearance (e.g., size and charge, or size and inactivation). Most mAb processes use a low pH hold for virus inactivation in combination with size-based virus filtration. Significant viral clearance can also be achieved in many chromatographic processes [38]. Viral inactivation is currently performed as a batch operation, but it should be possible to develop continuous inactivation processes using a tubular 'reactor' with a static mixer to achieve uniform residence time.

Genzyme has recently presented results for the production and initial purification of both a monoclonal antibody and a recombinant human enzyme using an integrated continuous process [39••]. This system used a perfusion bioreactor with an alternating tangential flow filtration (ATF) system for cell recycle, with the clarified cell culture fluid directly purified using a multicolumn periodic counter-current chromatography (PCC) system. The processes ran continuously for 30 days with good product quality. The continuous system had a much smaller equipment footprint and far greater productivity than a conventional batch process, demonstrating the technical feasibility of implementing an integrated continuous process for production of key biotherapeutics.

Challenges for continuous bioprocessing

Although the basic principles required for an integrated continuous bioprocess are reasonably well established, there is still considerable disagreement within the bioprocessing community as to whether this is really the appropriate future direction for biomanufacturing [40•,41]. Continuous processes are likely to be lower cost, but this needs to be balanced against the greater risk associated with the implementation of technologies that have yet to be proven in commercial operation. This is particularly true for many of the continuous chromatographic separations that are still under development. There are also questions about equipment robustness and operability, maintaining sterility over long processing times, challenges with start-up and shut-down, uncertainties about regulatory approval, and the time and cost associated with the development of continuous processes — speed of development is absolutely crucial in bringing new therapies to market.

There is also a critical need for development of real-time process analytical technology (PAT) that is capable of identifying changes in key product quality attributes (e.g., aggregation and glycosylation, among others) during continuous processing [42••]. This PAT will need to be integrated with appropriate global process control strategies to ensure that the continuous process remains within defined specifications. This includes coordination of flow rates, management of process interruptions, and adjustments in operating conditions due to degradation in performance (e.g., loss of flux due to membrane fouling or loss of resin capacity after multiple cycles). This type of global process control will require coordination between unit operations and between upstream and downstream processes, both of which are absent in current biomanufacturing due to the batch nature of existing processes.

Conclusions

Although fully continuous processes have yet to be employed for large-scale purification of therapeutic proteins, much of the needed technology is already in place: there is extensive experience with the use of continuous perfusion bioreactors for production of a wide range of recombinant proteins and there have been significant advances in the development of continuous unit operations for downstream processing. Continuous processes have the potential to provide significant reductions in manufacturing costs (both operating and capital), with greater flexibility in managing multiple products within a single facility. Just as significantly, the shorter residence time and more uniform microenvironment in continuous processes should translate into improvements in product quality, particularly if these can be effectively integrated with appropriate advances in real-time process analytical technology and global control strategies. Although it is highly unlikely that continuous processing will displace existing capital investments in large-scale batch manufacturing for well established companies in the near future, there are likely to be exciting opportunities for implementation of fully continuous or hybrid (partially continuous) processes for greenfield facilities designed for production of important biotherapeutics.

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